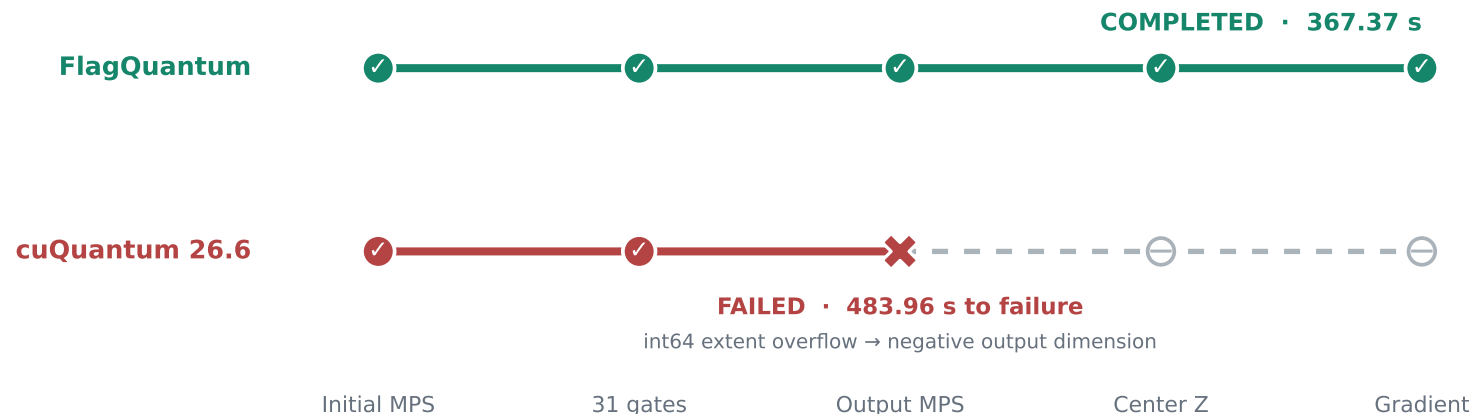


Same MPS workload: completion outcome

FlagQuantum completed the measured 131,072-qubit training step; cuQuantum 26.6 did not produce an output state.

131,072 qubits · $\chi=768$ · complex64 · 16 RY + 15 RXX · center-Z observable · one gradient step

a Measured execution path



b Execution context

FlagQuantum

2 nodes · 16×A800

Distributed MPS

cuQuantum NetworkState

1×A800 · device_id=0

Public MPS path is single-device

Same 131,072-qubit workload: FlagQuantum completed; cuQuantum 26.6 did not complete.

Completion/capability comparison—not a speedup claim. cuQuantum time is time-to-failure. Measured 2026-08-06 on NVIDIA A800 80 GB GPUs.